

Data Analysis Across Countries

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1. Introduction

As the world continues to progress, it dawns on us that we are living in an era of digitalized world where information is overloaded and where data plays a prominent role in making decisions and forming strategies. It is crucial to understand the role that data plays in our digitalized world. Organizations all around the world incorporate data-driven decisions in their strategic planning. Analyzing data and drawing meaningful conclusions from it is crucial, especially for national or global planning, as it aids in policymaking and the evaluation of country-wide statistics. One challenge that hinders data analysis is the lack of access to the refined version of data. Many scholars and researchers struggle to draw meaningful conclusions because the data is not readily available. Even when data is accessible, it often contains various discrepancies. There are many global organizations such as the World Bank, United Nations, Our World in Data, and Organization of Islamic Countries (OIC) that have a wealth of data and statistics about countries and it is also readily available. The data is available in a vast range of dimensions, such as demography, economy, politics, education, trade, business environment, health, climate, war and peace, crime, and so on. Such data helps in identifying global trends, conducting SWOT (Strengths, Weaknesses, Opportunities, Threats), and predicting future trajectories for these countries.

Despite the availability and accessibility of data from the above-mentioned sources, there is still a lack of user-friendly platforms that allow for the exploration of the data and extraction of meaningful insights. The need to build an interactive dashboard is prevalent as the existing dashboards and databases published online do not have an interactive interface and only provide the opportunity to look at the basic visualization and analysis. Moreover, there is not a specific platform or dashboard reserved for Muslim countries, hence making it difficult for researchers who want to conduct research on this particular region. Our project aims to address this gap by building a prototype to facilitate research for Muslim regions. The data was gathered from credible sources like Our World in Data, OIC, and Climate Action Dashboard. The data was cleaned and processed to build a dashboard and perform comparative analysis. The dashboard incorporates data from 56 Muslim countries on various dimensions and allows the user to extract meaningful conclusions from the visualizations.

2. Literature Review

Indicators for Muslim Countries:

The majority of the research articles for Muslim-majority countries revolve around the dimensions of economy and trade. For the analysis across boundaries economy and health are considered to be the most prominent indicators. Jani Kinnunen quantified the relationship within Islamic countries. The indicators were GDP per capita, index and sub-indices of economic freedom, and the growth rate of the countries over the years. Statistical analysis was used to analyze the data [5]. Moreover, regression analysis was used in one of the research articles to check whether values explain the low employment rates of Muslim women around the world [6].

Health indicators and associated factors were also analyzed across countries for Muslim-majority regions by Abdul Razzak. The research article highlights that statistical analysis was used to compare and analyze the data of Muslim-majority and non-Muslim-majority countries for 4 primary outcome measures i.e. male and female life expectancy, maternal mortality ratio, and infant mortality rate [7].

Wang investigated the causality between healthcare expenditure rise and economic growth for 31 countries by using both panel regression and quantile regression techniques over the period 1986-2007. According to the panel estimation results, Wang argues that while healthcare expenditure growth has a positive impact on economic growth economic growth decreases healthcare expenditure growth. However, the results of quantile regression indicate that the effect of healthcare expenditure growth on economic growth is positive in countries that have medium and high levels of economic growth [8].

There were also several researches conducted on the dimension of the business environment. A study predicted the Islamic stock market indices for the next 25 months by analyzing the data in 6 major Muslim countries employing statistical analysis [9]. Another study investigated whether sensations and emotions have an impact on market sentiment and price fluctuations by analyzing the data gathered from Twitter, search query of a list of words related to Islamic context, and the degree of engagement on social media [10].

The above literature review helped our project in shortlisting and finalizing the indicators and dimensions that should be added to our dashboard. Although the research on Muslim-majority countries as a group was limited, it still helped us shortlist the

dimensions we found suitable for our project.

3. Methodology

3.1. Research Design:

The research design for this study follows a structured data analysis approach to assess global trends and patterns across multiple dimensions for 56 Muslim-majority countries. The overall approach involves several key steps, including data collection, cleaning, consolidation, and the development of a data analysis dashboard.

Data Collection Methods:

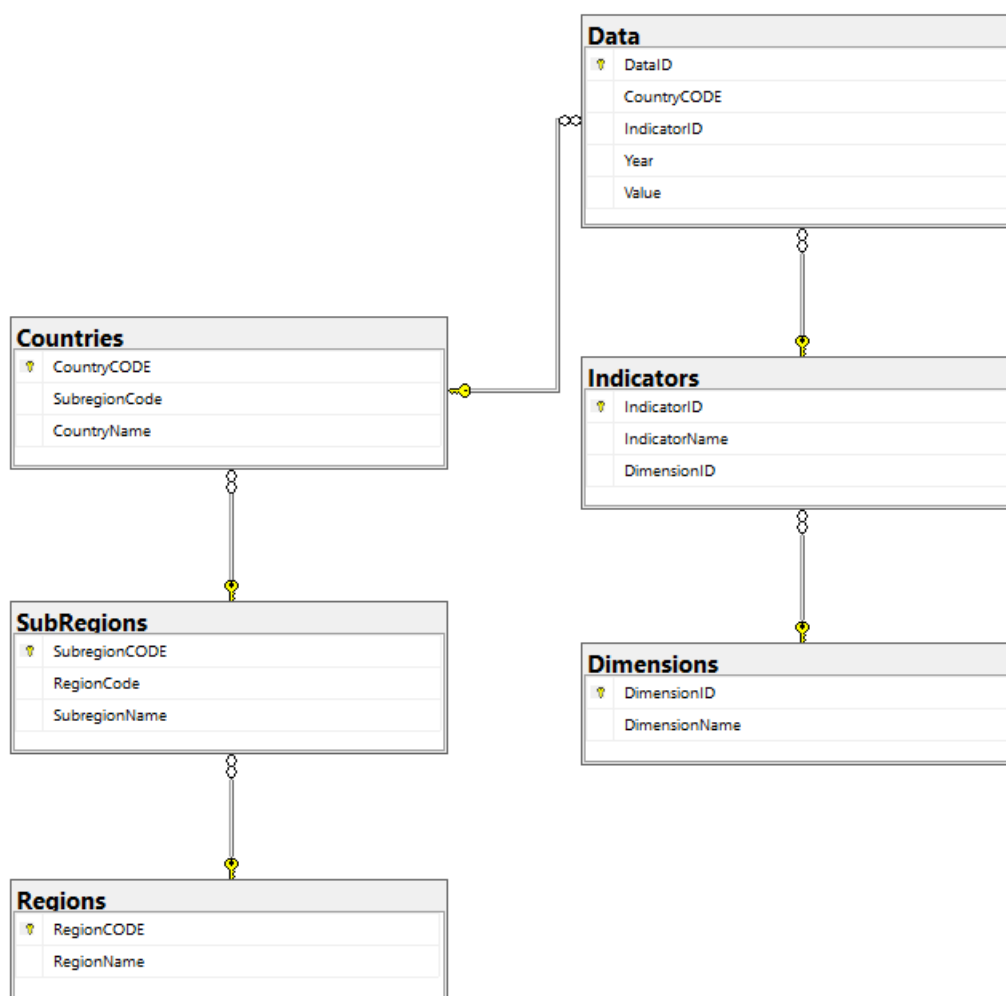
- **Primary Data Sources:** Data was collected from three credible online databases: [World Bank Open Data](#), [OIC Statistics \(OICStat\) Database](#), and [Climate Action Dashboard](#).
- **Dimensions:** The data encompasses various dimensions, including agriculture, business environment, climate and environment, development, economics, health, and technology & innovation.

Sample Selection:

- **Geographical Scope:** The study focuses on 56 Muslim-majority countries recognized by the Organization of Islamic Cooperation (OIC) across Asia, Africa, South America, and Europe.
- **Time Frame:** Data spans from the year 2000 to 2023.

Experimental Procedures:

- **Data Cleaning and Consolidation:** Utilizing Python scripts to clean and preprocess data before loading it into an MSSQL server database.
- **Database Design:** The database is structured into tables; regions, subregions, countries, dimensions, indicators, and data.



The chosen methods are appropriate for addressing the research problem as they ensure comprehensive data coverage, reliability from credible sources, and a robust database structure for in-depth analysis.

3.2. Data Collection:

The specific methods used to collect data for this research involved programmatic access and downloading from the identified online repositories:

- **World Bank Open Data:** Provided extensive country-specific data across various domains.
- **OIC Statistics (OICStat) Database:** Offered data on member countries of the Organization of Islamic Countries with 1852 indicators under 27 categories.
- **Climate Action Dashboard:** Supplied global climate and environment data.

These methods were chosen based on their credibility, comprehensiveness, and accessibility. Implementation involved:

- Programmatic Data Gathering:** There is a total of 57 countries listed in the OIC out of which 23 countries have declared Islam as their state religion. Numerous publicly available online databases are the primary sources of data for the analysis. The World Bank database is one of the most prominent sources of data as they have a vast range of dimensions with each having many different indicators [1]. The data is publicly accessible, and the interface of their website also allows the user to download pre-designed reports and look at basic visualization. However, the user cannot do a detailed analysis of the data across multiple dimensions. Another repository is the OIC Statistics (OIC Stats) database. It provides data on the member countries of the Organization of Islamic Countries and hosts 1852 indicators under 27 dimensions with data dating back to 1970. However, this website also does not have an interactive interface that allows the user more visualization options other than plotting some graphs [2]. Moreover, Our World in Data, an open-source platform, offers a vast repository of global data along with analytical reports addressing contemporary issues supported by tables and visualizations. The website also allows users to download data for any sort of custom implementation [3]. Additionally, Climate Action Dashboard is a prominent source of collecting climate data on the countries. It also allows the user to download data for cleaning, processing, and analysis and has a good number of indicators [4].
- Data Cleaning:** Addressing data formatting, consistency, missing values, redundancy, and anomalies to ensure data integrity.
- Data Consolidation:** Transforming heterogeneous data into a consistent format and loading it into a central MSSQL database.

The Python and SQL scripts used can be accessed using the following [link](#).

These methods align with the research objectives by ensuring the data is reliable, comprehensive, and suitable for detailed analysis across the selected dimensions and countries.

3.3. Dashboard Development:

The result of this research focused on developing an interactive dashboard. After finalizing the database, Power BI was chosen as the tool for creating this dashboard due to its robust data visualization capabilities, ease of integration with various data sources, and user-friendly interface. Other tools like Qlik Sense and Tableau were considered, but Power BI was ultimately selected. The setup involved installing Power BI Desktop, connecting to the MySQL database by providing the server details, and loading the tables into Power BI. The dashboard was then designed to include various interactive features and visualizations that enable users to explore and analyze the data comprehensively.

4. Research Results / Findings / Discussions

The research produced two significant outcomes: a clean and preprocessed database containing comprehensive performance indicators for Muslim countries and an interactive online dashboard for data exploration and visualization. The database, compiled from reliable sources such as international organizations, includes data on eight dimensions: Economy, Health, Business Environment, Demography, Technology & Innovation, Climate and Environment, and Agriculture. It contains over 70 indicators at both country and region levels and underwent thorough cleaning and preprocessing to ensure its accuracy and suitability for analysis.

The interactive dashboard, developed using Power BI, is designed to provide comprehensive insights into various dimensions of development for Muslim countries. It features eight pages, each dedicated to a specific dimension, including Economy, Health, Business Environment, Technology and Innovation, Climate and Environment, and Agriculture. The Overview/Landing Page presents key demographic statistics and an interactive map, allowing users to select regions, countries, and years, updating the statistics accordingly. The subsequent pages focus on specific dimensions, featuring relevant indicators and visualizations. The Correlation Page, which is the last page, enables users to explore dynamic correlations between different indicators using dropdown menus.

The dashboard's interactive features, such as slicers and filters, enhance the user experience by allowing customized analysis. It is accessible online, making it a valuable tool for researchers, policymakers, and stakeholders worldwide. This comprehensive resource can facilitate informed decision-making and provide deeper insights into the

development indicators of Muslim countries.

The dashboard can be accessed online using this [link](#).

5. Limitations

- This dashboard does not depict the data of the global Muslim population but rather the data of the recognized Muslim Nations. This is because the data for Palestine wasn't available in the data repositories we utilized other than the OIC, hence, to keep all indicators' data consistent Palestine isn't included in the database. Moreover, data from countries with large Muslim populations, such as India, could not be included as they are not recognized as a Muslim State.
- Although the time frame of our data is 2000 to 2023, data for some of the indicators is unavailable for recent years as well as for a few indicators data is missing for a few years, so on our dashboard for those countries the visuals indicate it as “blank” indicating that the current dashboard is not comprehensive and still a work in progress.

6. Conclusion

Our project employed a structured approach focusing on data collection, cleaning, consolidation, and the development of an interactive dashboard for 56 Muslim-majority countries. Data was gathered from credible sources such as World Bank Open data, OIC Statistics Database, and Climate Action Dashboard. The data was cleaned and consolidated into an MSSQL database. Power BI, a powerful tool with excellent features of data visualization and user-friendly interface, was used to develop an interactive dashboard. We collected data from the mentioned sources across various dimensions including health, economy, technology, business environment, agriculture, and others. The data was cleaned effectively and all of the inconsistencies, anomalies, and missing values were removed from the dataset. The dashboard effectively addresses the problem statement by filling the gap in available resources for analyzing data of Muslim-majority countries. Our dashboard effectively allows for exploration and analysis of data across various dimensions proving to be an essential tool for researchers and policymakers. The dashboard is interactive which allows the user to customize their analysis, leading to more informed decision-making based on nuanced data.

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